Choosing the right interest rate

A guide to the simulator accompanying:

Davis, L. E., & Gómez-Ramírez, L. (forthcoming). A model for teaching the macroeconomics of tariffs with global supply chains. *The Journal of Economic Education*.

1. Setup

In the simulator, a one-point change in the MP interest rate changes current output by -4 in the next period:

$$Y_{t+1} = Y_t - 4\,(i_t - i_{t-1});$$

a one-point change in the MP interest rate changes equilibrium output by $+1$ in the next period:

$$Y_{e,t+1} = Y_{e,t} + (i_t - i_{t-1});$$

and inflation follows this Phillips curve:

$$\pi_{t+1} = \pi_t + \beta\, GAP_{t+1}, \qquad GAP_t = Y_t - Y_{e,t}.$$

So each point of the interest rate moves next period's gap by 5.

Before the tariff shock ($t = -1$) the economy is at its medium-run equilibrium (MRE): inflation is at the target and the gap is zero, $\pi_{-1} = \pi^T$ and $GAP_{-1} = 0$.

Worked example (a run with the simulator): $\pi^T = 2$, $\beta = 1$, $i_{-1} = 5$, $Y_0 = 100$, $Y_{e,0} = 95$; hence $GAP_0 = 5$ and $\pi_0 = 7$.

A useful identity. Because inflation starts at the target (and $GAP_{-1} = 0$), its distance from the target equals β times the sum of all gaps since the shock:

$$\pi_t - \pi^T = \beta\,(GAP_0 + GAP_1 + \cdots + GAP_t).$$

Inflation is on target exactly when this sum is zero. The column "Sum of gaps" in the tables below is this running sum (it includes the current period's gap).

2. The exclusively right-choices path

With no mistakes, three decisions take the economy back to the MRE (where i^* denotes the right decision):

- $t = 0$: $i_0^* = i_{-1} + \dfrac{GAP_0 + GAP_0}{5} = i_{-1} + \dfrac{2\,GAP_0}{5}$. The gap must swing from GAP_0 to $-GAP_0$ in $t = 1$ (that is, GAP_1 must be $-GAP_0$) so that inflation returns to the target at $t = 1$.
- $t = 1$: $i_1^* = i_0^* + \dfrac{GAP_1}{5}$. Inflation is already on target in $t = 1$, so the gap in $t = 2$ should only be closed (that is, GAP_2 must be 0).
- $t = 2$: $i_2^* = i_1^*$. The economy is at the MRE: inflation has been at the target since $t = 1$ ($\pi_1 = \pi_2 = \pi^T$) and $GAP_2 = 0$, so, to preserve this situation in $t = 3$ and thereafter, the rate must be left unchanged.

In the example: $i_0^* = 5 + \frac{2(5)}{5} = 7,\ i_1^* = 7 - \frac{5}{5} = 6,\ i_2^* = 6.$

t	i chosen	Y	Y_e	GAP	π (%)	Sum of gaps	Right i (rule)
-2	5	100	100	0	2	—	—
-1	5	100	100	0	2	—	—
0	7	100	95	5	7	5	$5 + \frac{5+5}{5} = 7$
1	6	92	97	-5	2	0	$7 + \frac{-5+0}{5} = 6$
2	6	96	96	0	2	0	$6 + \frac{0+0}{5} = 6$

If the right-choices path is entered, the simulation will carry on (just repeating the same values as in $t = 2$) from $t = 3$ until $t = 7$ (these last five periods are not shown in the table above).

3. The general case: any mistakes, any number, in any order

It can be deduced that a single rule gives the right rate in every period, from wherever the economy is and whatever happened before: choose the rate that brings inflation back to the target next period. Using the identity above, it follows that:[1]

$$i_t^* = i_{t-1} + \frac{GAP_t + (GAP_0 + GAP_1 + \cdots + GAP_t)}{5}.$$

In words: add this period's gap to the running sum of all gaps so far (this period's included), divide by 5, and add the result to last period's rate. Note that GAP_t appears twice: once on its own and once inside the sum. β does not appear. When inflation is on target the sum is zero and the rule reduces to closing the gap. (Note that the three rules of the right-choices path are this rule applied to a path with no mistakes.)

Recovery pattern. Only the last mistake matters: every new mistake restarts the recovery. If the last mistake is made in period k, the rule gives three right decisions:

- $\boldsymbol{k+1}$: the rule brings inflation back to the target (in $k+2$);
- $\boldsymbol{k+2}$: the sum of gaps is zero again, so the rule just closes the gap (in $k+3$);
- $\boldsymbol{k+3}$: the rate is left unchanged, which keeps the economy at the MRE (the confirmation step).

The economy is therefore back at the MRE at $t = k+3$. This assumes that the zero lower bound does not bind during the recovery and that the simulation does not end early (see Section 4.2).

[1] The Appendix shows how the rule follows from the identity.

Example with two mistakes. Same inputs as above.[2] The user enters $i_0 = 6$ instead of 7 (first mistake), then enters the right $i_1 = 7$, then enters $i_2 = 5$ instead of 6 (second mistake). After each mistake the same rule gives the right rate, and the recovery restarts from the second one:

t	i chosen	Y	Y_e	GAP	π (%)	Sum of gaps	Right i (rule)	Decision
-2	5	100	100	0	2	—	—	Pre-tariff
-1	5	100	100	0	2	—	—	Pre-tariff
0	6	100	95	5	7	5	$5+\frac{5+5}{5}=7$	Mistake
1	7	96	96	0	7	5	$6+\frac{0+5}{5}=7$	Right
2	5	92	97	-5	2	0	$7+\frac{-5+0}{5}=6$	Mistake
3	7	100	95	5	7	5	$5+\frac{5+5}{5}=7$	Right
4	6	92	97	-5	2	0	$7+\frac{-5+0}{5}=6$	Right
5	6	96	96	0	2	0	$6+\frac{0+0}{5}=6$	Right

The two mistakes are shown in shaded rows. The first (entering $i_0 = 6$ instead of 7) closes the gap at $t = 1$ but leaves $\pi_1 = 7$, above the target; the rule then gives $i_1 = 7$, which brings inflation back to the target at $t = 2$. The second mistake (entering $i_2 = 5$ instead of 6) pushes it off again; the rule then gives $i_3 = 7$, $i_4 = 6$ and $i_5 = 6$, and the economy is back at the MRE at $t = 5 = 2+3$. After that, the simulation will carry on (just repeating the same values as in $t = 5$) from $t = 6$ until $t = 7$ (these last two periods are not shown in the table above).

4. Further notes on the simulator

4.1. The time horizon

Decisions are taken from $t = 0$ to $t = 6$. Therefore:

- If the last mistake is made at $t \leq 3$, the MRE can be reached and confirmed.
- If the last mistake is made at $t = 4$, the economy reaches the MRE exactly at $t = 7$, with no decision left to confirm it; the simulator still ends with "Simulation complete. The economy is at its medium-run equilibrium."
- If the last mistake is made at $t = 5$ or $t = 6$, there is not enough time to return to the MRE. [Of course, students and instructors can still use the model to work out the path back to the MRE in these cases, and beyond.]

[2] Recall: $\pi^T = 2$, $\beta = 1$, $i_{-1} = 5$, $Y_0 = 100$, $Y_{e,0} = 95$; hence $GAP_0 = 5$ and $\pi_0 = 7$.

4.2. Approximate right rates, the zero lower bound, $150 < i$, i that would imply negative Y or Y_e, and input bounds in the simulator

- Answers within 0.005 of the right rate (roughly, correct to two decimals) count as right; the simulator then records the exact right value.
- If the rule gives a negative rate, the zero lower bound binds: the right answer is $i = 0$, and the simulator explains that conventional monetary policy alone cannot stabilize the economy, with the following message:

  > "Right — but only in the sense that this is as low as the rate can go ($0 \leq i$).
  > But even at zero, the interest rate cannot stimulate the economy enough to bring inflation back to the target: conventional monetary policy alone cannot stabilize the economy."

- A wrong rate above 150, or one that would push current or equilibrium output below zero, is not accepted: the simulator asks for another rate.
- If the rule would require a rate above 150, or would push current or equilibrium output below zero, inflation can no longer be brought back in a single period, and the simulation ends with the following explanation (the part in brackets states which limit would be violated):

  > "Your earlier decisions have moved inflation so far from the target that it can no longer be brought back in a single period: doing so would require [an interest rate above 150 / output below zero / equilibrium output below zero / cutting the interest rate to zero, which would push equilibrium output below zero].
  > The simulation ends here. Please try again."

- Input bounds: $0 < \pi^T \leq 30$; $0.1 \leq \beta \leq 2$; $0 \leq i_{-1} \leq 100$; $55 \leq Y_0 \leq 145$; $55 \leq Y_{e,0} \leq 100$. Within these bounds, the exclusively right-choices path never violates any limit.

Appendix: How the rule follows from the identity

Step 1: finding the required next-period gap. Recall that the identity from Section 1 holds in every period, including $t+1$:

$$\pi_{t+1} - \pi^T = \beta\,(GAP_0 + GAP_1 + \cdots + GAP_t + GAP_{t+1}).$$

The goal is to bring inflation back to the target next period, so we set $\pi_{t+1} = \pi^T$, which makes the left side of the identity zero. Since $\beta > 0$, for the identity to hold the sum in parentheses must be zero as well (that is, $GAP_0 + GAP_1 + \cdots + GAP_t + GAP_{t+1} = 0$). It follows that

$$GAP_{t+1} = -(GAP_0 + GAP_1 + \cdots + GAP_t).$$

That is, next period's gap must cancel everything that has pushed inflation away from the target. This is also why β drops out of the rule.

Step 2: finding how the interest rate affects the next-period gap. From the model,

$$GAP_{t+1} = Y_{t+1} - Y_{e,t+1} = \big(Y_t - 4\,(i_t - i_{t-1})\big) - \big(Y_{e,t} + (i_t - i_{t-1})\big) = GAP_t - 5\,(i_t - i_{t-1}).$$

Step 3: finding the rate that delivers the required next-period gap. To find the right choice of the interest rate, we equate the two expressions above for GAP_{t+1},

$$GAP_t - 5\,(i_t - i_{t-1}) = -(GAP_0 + GAP_1 + \cdots + GAP_t),$$

and solve for i_t:

$$i_t^* = i_{t-1} + \frac{GAP_t + (GAP_0 + GAP_1 + \cdots + GAP_t)}{5}.$$